

Lame Duck by Primary: Effects of Electoral Incentives on U.S. House Representatives

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Abstract

How do electoral incentives motivate representatives to exert effort and shape policy positions? While reelection is widely recognized as representatives' primary motivation in democratic politics, accurately estimating its effects is challenging because representatives self-select into rerunning or retiring decisions. Leveraging legislators' lame duck status after primary elections, this article estimates how electoral incentives affect U.S. House Representatives. Compiling data from 1954 to 2022, I compare legislators who win primaries and remain motivated for reelection to those who lose and exit the race. Through regression discontinuity and difference-in-differences designs, I find that primary losers significantly reduce effort-related activities, as evidenced by decreased roll call participation, bill sponsorship, and voter communication. In contrast, the results show that legislators do not adjust policy positions for reelection; both winners and losers maintain consistent ideological and party-line adherence. These findings illuminate the scope of electoral incentives' effects and contribute to debates on electoral accountability and democratic representation.

Keywords: Electoral accountability, electoral incentives, legislative politics, U.S. House Representative

1 Introduction

Does the pursuit of reelection prompt representatives to work harder or shift their policy stances? Numerous scholars have emphasized the critical role of electoral incentives in democratic systems, arguing that the desire to secure reelection motivates officeholders to remain closely aligned with their constituents' interests and expectations. Elected officials may be motivated by goals other than reelection such as advancing their preferred policy agendas or heightening their standing within political parties, but the prevailing view holds that “[the electoral goal] must be the *proximate* goal of everyone, the goal that must be achieved repeatedly if other ends are to be entertained” (Mayhew, 2004). In this way, a drive for reelection serves as a fundamental mechanism of democratic accountability, ensuring political leaders' responsiveness to voter preferences and setting democratic governance apart from autocratic rule.

Despite the importance of electoral incentives in democratic politics, however, existing research offers conflicting theoretical perspectives and mixed empirical findings on what specific behaviors these incentives elicit from representatives. In theory, electoral incentives could influence a representative's behavior in opposing directions. Representatives may strive to meet voters' demands to secure reelection (Mayhew, 2004; Fenno, 1978). At the same time, they may (or may not) feel pressure from party leaders to act otherwise as they control crucial campaign and career resources (Rohde, 2010; Aldrich, 2008; Cox and McCubbins, 2005; Krehbiel, 1993). Scholars also divide over the extent to which voters remain informed about ongoing politics, thereby disagreeing over how strong the effects of electoral incentives could be (Achen and Bartels, 2017; Ansolabehere and Jones, 2010; Campbell et al., 1980).

A core challenge in empirically evaluating theoretical claims about the impact of electoral incentives lies in disentangling these incentives from legislators' strategic decisions about whether to seek reelection or retire. Analyses on electoral accountability often rely on comparison of the behavior of incumbents who run for another term with the behavior of those who step down. However, rerunning or retiring decisions are shaped by a range of

personal and political factors, making it difficult to attribute observed behavioral differences solely to the presence or absence of reelection motives. Such a difficulty has made research on electoral accountability largely rely on theoretical assumptions and correlational evidence where credible causal designs have been difficult to implement. While previous scholars have employed various strategies to address this selection bias, robust empirical evidence remains relatively limited – particularly in the context of legislators in highly salient political environments such as U.S. House Representatives.

This article addresses this empirical challenge in the context of the U.S. House of Representatives by exploiting the unique opportunity presented by incumbent legislators’ lame duck status after primary elections. By collecting and combining data from over 3,800 incumbent legislators and their primary election contests between 1954 and 2022, I compare the post-primary behavior of primary election winners, who remain motivated for the upcoming general election, with that of primary election losers, who are disqualified from running for reelection. Although relatively uncommon, a sufficiently large number of incumbent legislators either lose or face significant challenges in primary elections, offering valuable opportunities to credibly estimate the effects of electoral incentives. In particular, in comparing the primary winners and losers, they allow me to ensure their comparability through two complementary research designs: a regression discontinuity design, which compares incumbents who narrowly win or lose primary elections, and a difference-in-differences design, which contrasts legislators’ behavior during terms when they lost primaries with their behavior in terms when they secured renomination.

The findings from both designs provide strong evidence that electoral incentives play a significant role in determining the level of effort legislators invest. After losing primary elections, legislators show a notable decline in roll call participation, bill sponsorship, cosponsorship, and the frequency of e-newsletters sent to constituents. To further investigate whether it is the pressure of upcoming general elections that drives this behavior, I analyze how these effects vary with the expected competitiveness of each district’s general election. I find

that the effects are substantial mostly in districts where general elections are expected to be competitive. This suggests that electoral concerns, rather than non-electoral considerations such as the anticipation of serving another term or future committee assignments, are likely the primary drivers of the results.

On the other hand, I find that both primary election winners and losers remain largely consistent in terms of ideological extremity and party-line adherence following primary elections. Contrary to much of the theoretical literature (e.g., Downs, 1965), this suggests that either the presence or absence of electoral incentives does not trigger systematic shifts in legislators' policy positions. To investigate the extent of this stability, I further restrict the analysis to bills that receive greater attention from party leaders (e.g., whipped bills, as noted by Evans, 2012), bills that scholars have marked as significant (Curry and Lee, 2020), and contested bills that are close to passing or failing. The results show that neither primary election winners nor losers significantly change their policy positions even for these bills, underscoring the substantive strength of the null findings.

These findings carry several important implications. First, they evaluate the foundational assumptions in the literature on electoral accountability by delineating the scope and magnitude of electoral incentives' influence. The initial set of results – demonstrating that electoral incentives motivate legislators to increase effort-based activities – reinforces the standard logic of electoral accountability. In contrast, the second set of results – indicating that electoral incentives do not meaningfully affect legislators' policy positions – challenges the traditional Downsian framework of electoral competition and instead supports the view that legislators' ideological stances remain largely inflexible. These findings move the literature from largely theoretical accounts toward a more empirically grounded understanding.

Second, the finding that legislators' policy positions are barely affected by lame duck status contributes to other important debates in the literature. In particular, the finding suggests that electoral competition – or the lack thereof – may not be a primary driver of ideological polarization, contrary to existing explanations (McCarty, 2019; Barber and

McCarty, 2015; Burden, 2004). Instead, by showing that legislators’ ideological stances remain stable, the results indirectly suggest that the widening partisan and ideological divides are more likely due to electoral replacement of incumbent legislators than by incumbents themselves adjusting their policy positions in response to electoral pressures (Moskowitz, Rogowski and Snyder, 2024).

The findings also inform debates on party influence (Cox and McCubbins, 2007; Krehbiel, 1993) and the reliability of policy position measures based on roll call votes, such as DW-NOMINATE scores (Poole and Rosenthal, 1985; McCarty, 2016). They show that legislators adhere to party-line voting regardless of their lame-duck status, implying that electoral incentives do not serve as a meaningful channel for party control. This stability in roll-call behavior further suggests that these ideological measures primarily capture legislators’ enduring, intrinsic preferences rather than shifts driven by electoral concerns.

1.1 Theoretical Considerations on Legislators’ Behavior

To understand how electoral incentives affect legislators’ behavior, this article focuses on two key aspects: effort-related behavior and policy positions based on roll call votes. Standard models of electoral accountability suggest that electoral incentives drive elected officials to increase their effort levels in hopes of securing reelection (Barro, 1973; Ferejohn, 1986). However, when representatives must allocate limited resources across multiple tasks, it is less clear which specific activities will be prioritized (Fenno, 1973; Holmstrom and Milgrom, 1991). For example, although electoral incentives should theoretically motivate legislators to attend roll call votes to avoid perceptions of shirking, the competing demands of campaigning and district responsibilities may lead those focused on reelection to miss these votes more often. The prediction is further complicated by the possibility that voters are not sufficiently attentive to or informed about ongoing politics; as a result, legislators may adjust their behavior only in limited ways. Scholars remain divided over the extent of voter attentiveness and, consequently, the strength of electoral incentives (Achen and Bartels, 2017;

Ansolabehere and Jones, 2010; Campbell et al., 1980; Key and Cummings, 1966; Miller and Stokes, 1963).

When it comes to legislators' policy positions, many prominent models predict that electoral incentives pull legislators toward the median voter's position (Downs, 1965; Calvert, 1985; Wittman, 1977, 1983). If Democratic legislators are positioned to the left of their district's median voter and Republicans to the right (Bafumi and Herron, 2010), these theoretical models predicts that primary winners will shift their positions toward the center, while primary losers may not exhibit the same moderation. On the other hand, some scholars suggest that electoral incentives may drive legislators to take more partisan positions because it may help mobilize their supporters (Adams and Merrill, 2003; Fiorina, 1974; Huntington, 1950). If loyal voters' turnout decisions are more sensitive to their candidates' positions than median voters are, then candidates could benefit from shifting toward positions that more strongly appeal to their core supporters.

Another class of theoretical models, including citizen-candidate models, makes a starkly contrasting assumption: that legislators' policy positions remain largely fixed (Alesina, 1988; Osborne and Slivinski, 1996; Besley and Coate, 1997). One interpretation is that political candidates are ideologues who steadfastly adhere to their preferred ideological stances. Alternatively, the fixed policy positions can be understood as an assumption that candidates are perceived by voters based on their inherent ideal positions, regardless of the policy stances they publicly adopt. According to this strand of models, electoral incentives have no effect on legislators' policy positions.

It also remains unclear how much a party influences its legislators' behavior through electoral incentives. Despite continued research, it has remained difficult to identify the party influence because of estimation challenges, including legislators' selection into parties (Smith, 2007; Krehbiel, 1993; Cox and McCubbins, 2007). If party influence exists, it is plausible that legislators adopt party-line stances when seeking reelection, as the party may supply essential resources for their campaigns. It is also possible that legislators' behavior is

affected by various forms of interest group influences, although the existing literature lacks substantial empirical evidence demonstrating the significant influence of political donors and interest groups (Ansolabehere, de Figueiredo and Snyder, 2003; Fowler, Garro and Spenkuch, 2020). Given that these diverse considerations may pull legislators in different directions, it is unclear how electoral incentives would affect a legislator’s policy position.

When comparing legislators who seek reelection with those who retire, it is crucial to account for the motivations and incentives facing retiring legislators. Although a common assumption is that legislators freed from the demands of reelection may reduce their legislative activity or coast through their remaining term, they can also be driven by alternative motivations. For instance, they may endeavor to maintain a favorable standing within their party for future electoral bids or leadership roles, or they may prepare for post-congressional careers – such as lobbying, think tank fellowships, or executive-branch appointments – and adjust their engagement accordingly (Shepherd and You, 2020; Jenkins and Nokken, 2008a)¹. Personal policy preferences or ideological commitments may also keep them active regardless of electoral pressures. Depending on the specific research design and context, legislators who forgo reelection – due to a term limit or a primary defeat – may therefore remain active in Congress, not for immediate voter approval, but to advance longer-term career ambitions or personal priorities.

1.2 Previous Research on Electoral Incentives

Empirically estimating the effects of electoral incentives is fundamentally challenging because it involves addressing a counterfactual question. To understand how electoral incentives influence a legislator’s behavior, we need information on what the legislator would do both in the presence and absence of these incentives. However, one of these scenarios is inher-

¹In the appendix, I examine the potential effects of anticipation of the president’s appointment by restricting the analysis to legislators in midterm years who belong to the president’s party; these lawmakers are more likely than others to anticipate a post-Congress appointment since the current president is certain to serve at least two more years (Table A.5). The analysis does not provide strong evidence that the anticipation of appointment significantly shapes losing legislators’ behavior.

ently counterfactual and thus unobservable, preventing researchers from directly measuring how electoral incentives change a legislator’s behavior. The key inferential challenge lies in identifying a credible method to estimate a legislator’s counterfactual behavior.

To understand the impact of electoral incentives, earlier literature on shirking examined whether legislators behave differently in the last congressional terms when they voluntarily retire (Lott, 1987; Lott and Davis, 1992; Zupan, 1990). Their idea was to employ the fact that unlike rerunning legislators, voluntarily leaving legislators do not face electoral incentives. However, a rigorous analysis requires that staying and leaving legislators are comparable. Specifically, the behavior of legislators who leave should reflect what staying legislators would have done if they had left, and vice versa. If legislators make strategic decisions about whether to seek reelection or retire, or if, for example, those who retire are of lower quality or hold more extreme views than those who seek for reelection, simply comparing the two legislator groups can introduce significant selection bias.

Building on earlier research, scholars have attempted to address these empirical concerns by employing various research designs to disentangle the effects of electoral incentives from those of electoral selection. These approaches include leveraging term limits (Alt, Bueno de Mesquita and Rose, 2011; Fourniaies and Hall, 2022; Besley and Case, 1995), electoral reforms (Motolinia, 2021), and parties’ candidate nomination processes (Fiva and Nedregård, 2024).

Similar to this article, Fiva and Nedregård (2024) employ party renomination decisions in Norway to compare legislators’ within-term behavior following the renomination outcome. Both studies find that legislators who fail to secure renomination tend to reduce their effort-related activities while largely maintaining their policy positions. Aside from questions about whether renominated and non-renominated legislators are held comparable, the two works are situated in markedly different political contexts. In Norway, party leaders decide which incumbent legislators are renominated, a process characterized by strong party discipline and limited personal campaigning. In contrast, in the United States, incumbents must compete in primary elections that emphasize individual candidacies. Consequently, while Fiva and

Table 1: Career Trajectory After Losing in Primary Election

Run for Elections Later	Non-elected Public Position	Non-public Career	Retire or No Records
28 (17.50%)	27 (16.88%)	63 (39.38%)	42 (26.25%)

Note: The table shows the distribution of House representatives’ post-Congress career trajectories after losing in primary elections. Only 17.5% of them ended up running for an electoral position again, while the majority pursued non-electoral or non-public roles, retired, or left no trace of their activities after leaving Congress.

Nedregård (2024) highlight the influence of party leadership in shaping legislators’ behavior, this article centers on how individual legislators pursue reelection under a more candidate-driven electoral system.

Using a term-limit design with state legislators, Fourniaies and Hall (2022) report findings similar to those in this article: once state legislators reach their term limits and are no longer eligible for reelection, their overall productivity declines, yet their policy positions remain largely unchanged. The key difference from this article lies in the salience of the elections. U.S. House races typically attract far more public attention than state-level elections. Past research largely attributes the negligible effect of electoral incentives on state legislators’ policy positions to the low-salience setting of state-level contests (Rogers, 2017, 2023; Fourniaies and Hall, 2022). Hence, while it is surprising that electoral incentives affect state legislators’ productivity in a relatively low-salience environment, the findings in this article are striking in revealing that, even in the high-salience context of a U.S. House race, electoral incentives do not significantly alter Representatives’ policy positions – contrary to claims from previous studies (e.g., Ansolabehere, Snyder and Stewart, 2001; Griffin, 2006; Fiorina, 1973).

Compared to a term-limit framework, this article’s use of primary elections offers a more precise way to differentiate between legislators motivated by electoral considerations and those who are not. As most House Representatives exit electoral politics after a primary defeat (Table 1), comparing primary winners and losers provides a clearer gauge of the impact

of electoral incentives on legislative behavior. This contrasts with research on term-limited state legislators or governors, who often pursue other political roles, making it challenging to isolate the direct effect of electoral incentives.

Indeed, as Fourniaies and Hall (2022) and Olson and Rogowski (2020) highlight, termed-out state legislators may still be driven by future political prospects and party pressure. Fourniaies and Hall (2022) provides indirect evidence of this, finding a more significant negative effect of electoral incentives in states like California and Oklahoma where term limits prevent running for the other chamber. This suggests that when term limits don't preclude other candidacies, termed-out legislators may continue strategic behavior anticipating future opportunities. Similarly, Olson and Rogowski (2020) show that even after hitting term limits, state legislators continue to face considerable party pressure. These ongoing incentives, likely tied to broader political career ambitions, can substitute for direct reelection motivation and thus complicate the isolation of behavior specifically driven by immediate electoral incentives.

Several past studies employ a similar approach to this article's design by examining House Representatives' behavior during the lame duck sessions that follow general elections and precede the start of the next term (Jenkins and Nokken, 2008*a,b*; Schönenberger, 2024). They compare legislators who win reelection in general election and thus expect to remain in office to other legislators who lose and consequently expect to leave soon at the end of their current terms. Such designs yield credible estimates of electoral incentives by measuring the changes in the same legislators' behavior over a single term, effectively controlling for personal characteristics. Aside from whether winners and losers are held comparable, however, this approach encounters a few limitations. lame duck sessions are typically brief and involve few substantive legislative agendas, leading some researchers to rely on historical periods before the 20th Amendment in 1933, which significantly shortened the length of lame duck sessions. Moreover, party leaders sometimes engage in strategic agenda-setting during lame duck sessions – anticipating imminent shifts in Congress's partisan composition – which complicates determining the extent to which legislators' behavior is driven solely by reelection

concerns.

Another significant challenge in interpreting findings from lame duck session designs lies in the uncertain motivations of general election winners during this period. With a full two-year term ahead, these winners are less immediately focused on reelection than primary winners. Consequently, they may prioritize longer-term goals like securing committee assignments or advancing policy, making it difficult to discern if observed differences between general election winners and losers stem from electoral incentives or other factors. In contrast, primary winners face a general election within months, making reelection their most likely primary motivation and thus offering a clearer basis for assessing the effects of electoral incentives.

Finally, Cowburn and Sältzer (2025) employ a research design most similar to this article’s by comparing winners and losers of primary elections among U.S. House representatives. They analyze legislators’ language usage on X (Twitter) in 2020 and find that Democratic primary losers took significantly more moderate positions after their defeat, whereas no such effects were observed among Republicans. This article examines a wider array of legislative behavior – focusing on various effort-related activities and policy positions based on roll call votes – over a more extensive period from 1954 to 2022.

2 Data on Primary Elections and Legislator Behavior

In the United States, primary elections typically occur several months before the general election, providing parties an opportunity to select their preferred candidate to compete for the office. For incumbent legislators, these contests are especially significant: a primary win maintains their eligibility and drive to compete in the general election, while a defeat effectively disqualifies them from running for reelection.

This article utilizes these features of primary elections and examines how electoral incentives shape the behavior of House representatives by comparing incumbent House rep-

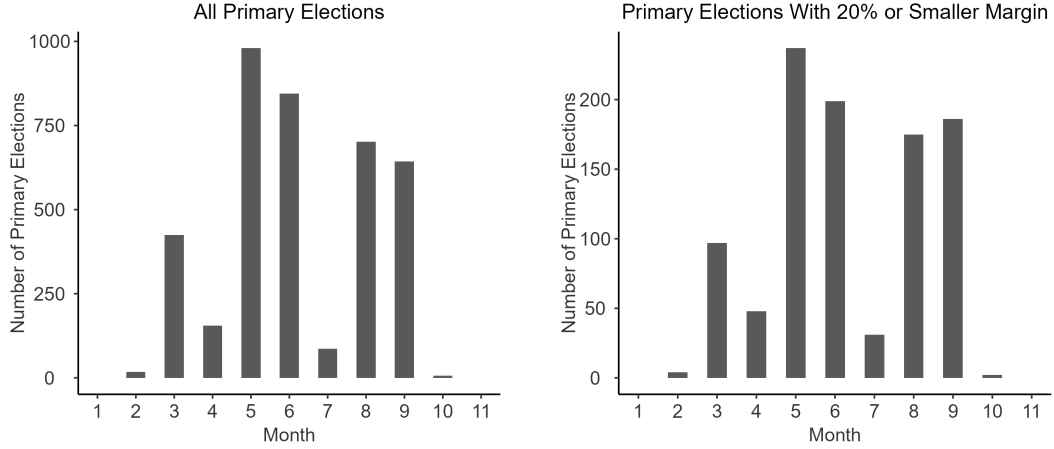
representatives who win their primary elections to those who lose. Specifically, I compare the post-primary behavior of primary winners – who remain eligible and motivated for the upcoming general election – to that of primary losers – who are effectively disqualified from seeking reelection.

To implement the idea, I began by collecting data on past primary elections. Pettigrew, Owen and Wanless (2014) provide primary election results for U.S. House Representatives from 1956 to 2010, and Miller and Camberg (2020) provide them from 2012 to 2018. These data do not include multiple cases of runoff elections, and I manually collected them from state archives and past newspaper articles. I collected additional primary election results for 2020 and 2022 from each state’s archives. The combined data cover U.S. House Representatives from 84th to 117th Congress, or 1954 to 2022, a total of 14,542 Congress-legislator units, and 2,668 distinct legislators. Of them, I focus on 3,860 Congress-legislators (1,546 distinct legislators) who competed in primary elections with at least one another challenger. There were 3,700 victories and 160 defeats.

In analyzing legislators’ policy positions, I focus on the patterns of their roll call votes, including ideological positions on liberal-conservative scale as well as their party-line stances. For legislators’ effort-related behavior, I examine key legislative activities that serve as proxies for a legislator’s effort in general, drawing on previous literature (Dal Bo and Rossi, 2011; Titunik, 2016; Fourniaies and Hall, 2022). These activities include roll call participation, bill sponsorship and cosponsorship, and the frequency with which their e-newsletters are sent to constituents.

The House representatives’ information and roll call voting records are collected from [voteview.com](https://www.voteview.com) (Lewis et al., 2024). Sponsorship data come from the Congressional Bills Project and cover from 1973 to 2018 (Adler and Wilkerson, 2020), and James Fowler’s cosponsorship network data provide dates of each cosponsorship from 1979 to 2004 (Fowler, 2006). E-newsletter data come from DCinbox (Cormack, 2024) and cover from 2010 to 2022. Because the legislators’ general election results may affect their electoral incentives, I did

Figure 1: Monthly Distribution of Primary Elections



Note: The left figure shows the month distribution of all primary elections, while the right figure shows those with less than 20% point margins. In both cases, June was the median month.

not include in the analyses legislators’ activities after general election dates.

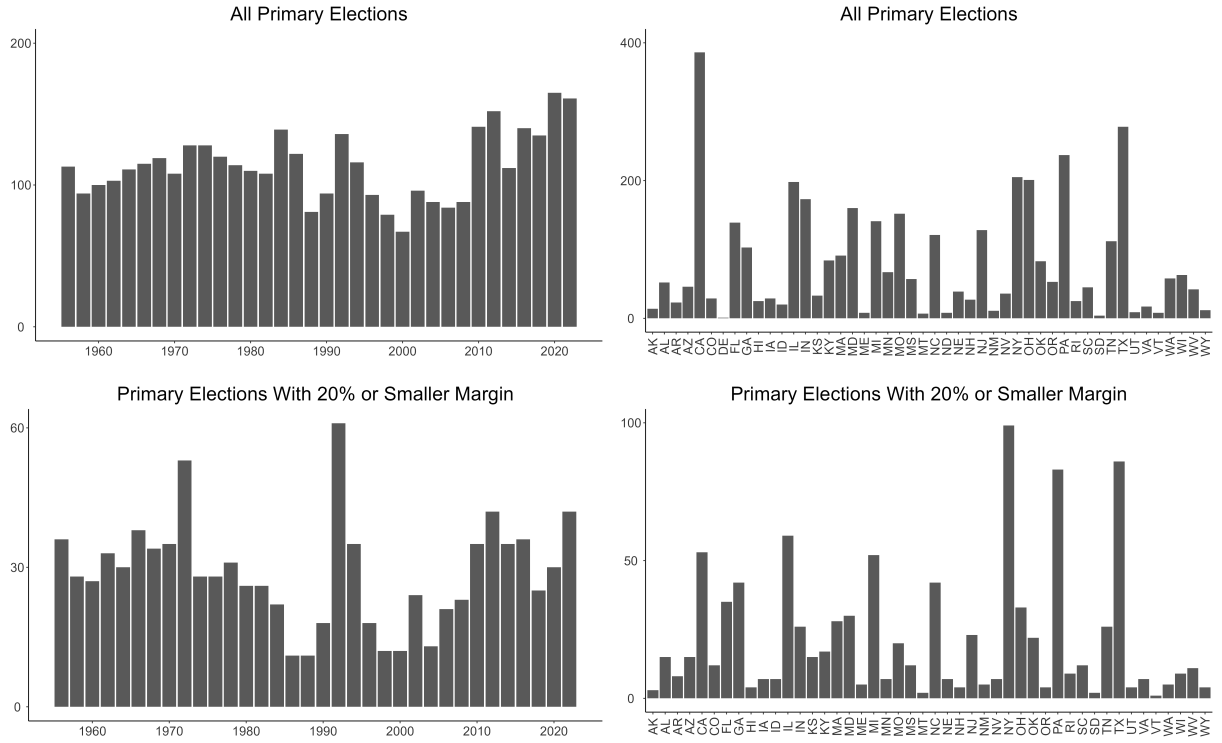
Since the key idea of this article is to examine legislators’ behavior after the primary elections, it is necessary to know the exact dates of primary elections. Federal Election Commission provides the entire states’ primary election dates back to 1996 and a few state archives provide earlier primary election dates. For most of the other years and states back to 1956, I manually collected the dates by accessing past newspapers.

Figure 1 shows the distribution of primary election dates by month, and Figure 2 shows the distribution across states and time. The median month of primary elections was June of the general election year. Total three million roll call votes were cast by the legislators in the analysis, and about 22 percent of the votes were cast after each legislator’s primary election date.

3 Empirical Strategy

To examine the behavior of primary winners and losers, I first use a regression discontinuity design, comparing legislators who narrowly won versus narrowly lost close primary races. I also employ a difference-in-differences approach to track how legislators’ behavior changes

Figure 2: Distribution of Primary Elections by Year and State



Note: The graphs on the left and right illustrate the number of primary elections by year and by state, respectively. The top rows display the distribution for all primary elections, while the bottom rows show the distribution for those with margins of less than 20%.

across terms in which they won primaries versus terms in which they lost. Both designs share the goal of ensuring that primary winners and losers are as comparable as possible: the regression discontinuity design achieves this by exploiting the as-if random outcomes of close primary elections, while the difference-in-differences approach does so by analyzing same legislators' changes in behavior along their career. Each design has its own strengths and weaknesses, and together they provide more robust evidence on the impact of electoral incentives on legislative behavior.

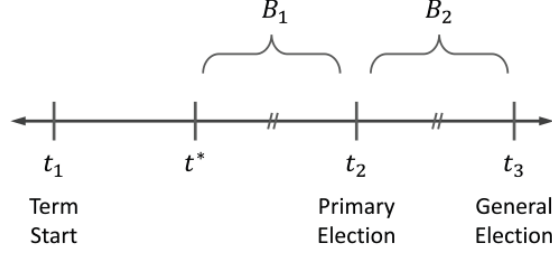
3.1 Measurement of Legislator Behavior

Before detailing the research designs, I first outline the key variables in this study. The unit of analysis is an individual House representative in a given Congress. The main independent variable distinguishes whether an incumbent House member wins or loses a primary election, capturing the presence or absence of electoral incentives for the general election. The dependent variable then tracks how legislators behave after the primary, reflecting their response to the electoral incentives. To account for each legislator’s baseline behavior, I use legislators’ within-term change in behavior from pre-primary to post-primary periods. Using within-term change accounts for varying base levels of legislators’ behavior as well as their potential behavioral trend made in anticipation of the primary, thereby isolating each legislator’s reaction to primary outcomes.

To analyze behaviors such as the number of bills sponsored, I construct a dependent variable that reflects the proportion of these activities conducted after the primary election. For example, if a legislator sponsors six bills before the primary and four after, the dependent variable is calculated as the proportion of post-primary sponsorship, or $\frac{4}{6+4} = 0.4$. Since primary election dates vary by state and year, a simple comparison of the amounts of pre- and post-primary activities may not be appropriate; legislators in states with later primary dates naturally have shorter time for post-primary activities. To address this, I introduce a reference date t^* for each legislator-year, which is set earlier than the primary date so that the time length between t^* and the primary is equal to the time length between the primary and the general election (Figure 3). Then I measure post-primary activity B_2 between the primary date (t_2) and the general election date (t_3), and pre-primary activity B_1 between the reference date (t^*) and the primary date (t_2). The change in behavior is then measured by the proportion of post-primary activities, or

$$\frac{B_2}{B_1 + B_2}, \tag{1}$$

Figure 3: Dependent Variables for Effort-related Behavior



Note: This figure explains how I measure legislators' change in effort-related behavior after primary elections. A reference date t^* is set such that the length of period between t^* and the primary election date t_2 is same as that between the primary election date t_2 and the general election date t_3 . B_1 and B_2 measure a legislator's behavior in period between t^* and t_2 , and t_2 and t_3 , respectively. Then the dependent variable measures a legislator's change in behavior after primary election by $\frac{B_2}{B_1+B_2}$.

whose value higher than 0.5 signifies greater engagement in that behavior after the primary election. With this method, I measure legislators' numbers of bill sponsorship, co-sponsorship, and the number of e-newsletters sent to their constituents.

In measuring changes in legislators' roll call voting patterns, I construct the dependent variable using an OLS regression with fixed effects. Specifically, for each Congress t , I estimate the following model with fixed effects at both legislator and bill-party levels:

$$\text{Vote}_{ipbt} = \gamma_{it} + \delta_{pb} + \beta_{it}\text{AfterPrimary}_{ibt} + \varepsilon_{ipbt}. \quad (2)$$

In this model, the left hand side, Vote_{ipbt} , is a binary indicator set to 1 if legislator i from party p exhibits a specific voting behavior – such as participating in roll call voting – on bill b in Congress t , and 0 otherwise. The term γ_{it} represents legislator fixed effect and δ_{pb} accounts for bill-by-party fixed effects. The key variable, $\text{AfterPrimary}_{ibt}$, is binary indicator that takes the value 1 if legislator i faces the roll call for bill b after the primary election and 0 otherwise.

In this specification, the coefficient β_{it} represents the change in legislator i 's voting behavior in Congress t following the primary election, while controlling for legislator-specific

and bill-specific factors. Essentially, if a legislator’s voting pattern shifts after the primary, β_{it} captures the direction and the magnitude of that change. I use this coefficient as the dependent variable in my research designs to measure how legislators’ roll call votes change post-primary. I extend this approach to other aspects of roll call voting behavior by re-estimating the regression with alternative definitions of Vote_{ipbt} , such as aligning with the party majority, or voting in line with party leaders.²

I also measure legislators’ change in ideological extremism in roll call voting after primary elections. To do this, I coded each bill as either conservative or liberal using an algorithm based on legislators’ voting records and party affiliations (Fowler and Hall, 2012). For each roll call vote cast, I classify it as liberal if the legislator votes “yes” on a liberal bill or “no” on a conservative bill, and conversely, conservative if they votes “yes” on a conservative bill or “no” on a liberal bill. Then, the coefficient β_{it} in Equation 2 represents the change in a legislator’s tendency to vote conservatively after the primary election. Because roll call extremism corresponds to opposite directions on the ideological spectrum for Republican and Democratic legislators, I adjust the measure by multiplying β_{it} by -1 for Democratic legislators. Then, a legislator i ’s change in roll call extremism after primary election in Congress t is characterized by

$$\text{Roll Call Extremism}_{it} = \begin{cases} \beta_{it} & \text{if } i \text{ is Republican in Congress } t, \\ -\beta_{it} & \text{if } i \text{ is Democrat in Congress } t, \end{cases}$$

where β_{it} represents legislator i in Congress t ’s change in conservative voting after primary election. A higher value of $\text{Roll Call Extremism}_{it}$ indicates increased conservatism among Republicans or increased liberalism among Democrats. Additionally, I examine the absolute values of these changes to assess which legislators more frequently adjust their positions in either direction following primary election outcomes.

²When measuring the policy positions based on roll call votes, I follow existing measurement methods and include roll call votes where legislators voted either “yea” or “nay,” excluding any abstained votes, to ensure that the policy position estimates are not conflated with abstention behavior.

The primary reason for measuring legislators’ roll call voting in this specific way is to account for the possibility that the types of bills introduced before and after major primary election dates may differ systematically. By explicitly controlling for bill attributes, this approach ensures that any observed changes in legislators’ actions reflect their post-primary lame duck status separately from variations in the legislative agenda. Nevertheless, to ensure that the results are not driven by this specific estimation method, in the appendix, I also measure changes in legislators’ policy positions using an alternative method by calculating the difference of the average voting patterns between the pre- and post-primary periods, and show that the results remain consistent (Table A.6). In the appendix, I also conduct a descriptive comparison of bills over time, examining the distribution of topics of bills before and after primary elections (Figure A.3), as well as factors such as the weekly number of bills, the frequency of participation, proximity to the passing threshold (Figure A.4). The comparison does not indicate notable, systematic differences in bill topics and types before and after primary elections.

3.2 Research Designs

With the above measurements, I employ research designs that use the fact that an incumbent U.S. House Representative must either win the party’s primary election or be a sole candidate to advance to the general election and compete for reelection. While most incumbent House Representatives secure primary election victories with relative ease, this is not always the case. Occasionally, a prominent local figure’s entry into politics poses a threat to the incumbent. At other times, primary voters view their incumbent politician as old and incompetent, opting to replace them with a new candidate. Changes in district boundaries can also jeopardize an incumbent’s electoral prospects, as they may move loyal supporters to other districts. Consequently, some primary elections result in either the incumbent losing or facing a closely contested race. These variations in primary election outcomes provide the basis for estimating the effects of electoral incentives in this article’s research designs.

3.2.1 Regression Discontinuity Design

First, I implement the regression discontinuity design. In the design, the running variable V_{it} is a legislator i 's share of the top two candidates' vote less 0.5 in Congress t . For example, if an incumbent received 60% of the votes, and two challengers received 35% and 5%, respectively, the running variable is calculated as $\frac{60\%}{60\%+35\%} - 0.5 = 0.132$. This indicates that incumbents with positive running variables won primary elections, whereas those with negative running variables lost.

The construction of this running variable is not feasible in states with different primary election systems, and I exclude these states. Specifically, I exclude Georgia before 1962 due to its use of the county unit system for state elections, which operated similarly to today's electoral college system for presidential elections. I also exclude states with nonpartisan blanket primaries, such as the top-two or top-four systems used in Washington (since 2008), California (since 2012), and Alaska (since 2022). Additionally, I exclude Connecticut and Louisiana because their primary systems differ significantly; in Connecticut, party nominees are typically chosen at party conventions, making primaries rarely competitive, while Louisiana uses a nonpartisan blanket primary where winners can be elected outright.

I estimate equations of the standard regression discontinuity specifications,

$$Y_{it} = \beta_0 + \beta_1 \mathbb{1}\{V_{it} > 0.5\} + f(V_{it}) + \varepsilon_{it}. \quad (3)$$

As previously defined, V_{it} is the running variable, or a legislator's top-two winning margin in the primary election, and the dependent variable, Y_{it} , captures the legislator's change in behavior following the primary election. The function $f(\cdot)$ is a flexible function of the running variable, though, given the small sample size, I primarily employ a linear specification (Gelman and Imbens, 2019) with a bandwidth of 20 percentage points. In the appendix, I demonstrate that the main results remain mostly robust under alternative bandwidth selections (Figure A.2). The coefficient of interest, β_1 , represents the RD estimator, capturing

Table 2: Observable Characteristics of Legislators by Electoral Margin in Primaries

	(1)	(2)	(3)	(4)
	Close	Close	Close	Non-close
	Losers	Winners	Primaries	Primaries
N	159	820	979	2,881
Congress	100.08 (10.31)	100.16 (10.39)	100.14 (10.37)	101.34 (10.04)
Seniority	5.11 (3.31)	5.04 (4.01)	5.05 (3.90)	5.06 (3.90)
Age	57.99 (10.98)	56.27 (10.97)	56.55 (10.98)	55.40 (11.01)
Democratic party	0.65	0.62	0.63	0.62
Previous vote share in general	0.68 (0.15)	0.71 (0.16)	0.70 (0.15)	0.70 (0.15)
DW-NOMINATE score	-0.08 (0.40)	-0.05 (0.39)	-0.05 (0.40)	-0.06 (0.40)
DW-NOMINATE $\times (-\mathbb{1}\{\text{Democrat}\})$	0.37 (0.18)	0.35 (0.19)	0.35 (0.19)	0.37 (0.17)
Sponsored bills pre-primary	3.53 (3.75)	3.54 (4.42)	3.54 (4.32)	3.34 (4.02)
Cosponsored bills before primary	66.26 (42.63)	50.23 (39.22)	52.85 (40.17)	55.41 (44.90)
E-newsletters before primary	26.34 (40.30)	16.92 (16.38)	18.36 (21.90)	12.00 (11.08)
Roll-call participation before primary	0.90 (0.10)	0.91 (0.08)	0.91 (0.08)	0.94 (0.07)
Voting with party majority before primary	0.87 (0.10)	0.87 (0.10)	0.87 (0.10)	0.89 (0.09)
Voting with party leaders before primary	0.88 (0.10)	0.88 (0.11)	0.88 (0.11)	0.90 (0.09)

Note: Close primaries are those with 20% percentage point or smaller margins. Standard deviations are in the parentheses.

the effect of primary election victory on changes in legislators' behavior.

Columns (1) and (2) of Table 2 present a balance check between winners and losers in close primary elections within a 20-percentage-point vote margin. Both groups display similar characteristics, including age, seniority, as well as their legislative activities before the primary election.

3.2.2 External Validity of Regression Discontinuity Design

Before proceeding to the difference-in-differences design, I would like to briefly discuss the external validity of the regression discontinuity design. The regression discontinuity design results are derived exclusively from legislators who faced competitive primary elections,

which may limit their generalizability to the population of interest, or the broader legislative representatives. This article addresses this constraint by complementing the analysis with a difference-in-differences approach that leverages the full sample and deriving consistent results. However, following considerations suggest that findings from competitive primary elections may offer ways to reasonably extrapolate to the behavior of legislators in general, including those facing minimal or no primary challenges.

First, I compare the descriptive characteristics of representatives in competitive primary elections with those in uncompetitive ones and find no significant differences between the two groups. Columns (3) and (4) of Table 2 present legislators’ characteristics and behavior before the primary elections, distinguishing between those from close and non-close primary contests. Overall, the two groups appear similar across most observable variables, with the number of e-newsletters being the only statistically significant difference, as determined by a t-test at the 95% confidence level.

Second, a qualitative comparison of the reasons behind primary challenges suggests that close and non-close primary contests are not markedly different. Cowburn (2024) classifies the motivations for primary challenges among U.S. House Representatives from 2006 to 2020 by examining candidates’ campaign websites and media interviews, following the framework by Boatright (2014). By analyzing his data, I find that the underlying reasons for primary contests are largely similar for legislators facing close primary elections and those in less competitive races. Among close primary contests within 20% point margin, the most common reason for a challenge is the perception that the incumbent is too moderate (56%), followed by concerns over age and competence (17%), the incumbent being seen as too extreme (13%), scandals (5%), and national issues (4%). This distribution is not dramatically different from that of non-competitive primaries, where the most frequently cited reason remains the incumbent’s moderate ideology (45%), followed by age and competence (24%), the incumbent being perceived as too extreme (14%), national issues (4%), and scandals (0.7%).

Third, many close primary elections occur as a consequence of redistricting. Redistricting

can unexpectedly disrupt an incumbent’s electoral advantages by forcing incumbents to run against one another or by splitting their loyal voter bases across districts. Because these disruptions are not caused by the legislators’ own qualities or strategic decisions, close primary contests in these instances are not simply the product of systematic differences between legislators. To the extent a significant primary challenge arises from such an unforeseen external shock, the competitive dynamics observed in these close elections can reasonably be generalized to more typical legislators who face little or no primary competition.

For example, following the 2010 Census, Pennsylvania happened to lose a congressional seat while its state legislature simultaneously shifted from Democratic to Republican control. The newly Republican-led legislature redrew district boundaries, merging the districts of two incumbent Democratic representatives, Jason Altmire and Mark Critz, into a single district. As a result, the two incumbent Democrats had to compete in the primary election, and Altmire was narrowly defeated to Critz with a 49% to 51% margin. The data confirm that redistricting frequently precedes substantial primary challenges. On average, 40.14 House representatives face a primary election decided within a 20 % point margin during a redistricting year, compared to 25.85 legislators in a non-redistricting year. In a robust check, I restrict the analyses to only redistricting years and the results show consistent findings (Table A.2).

3.2.3 Difference-in-Differences Design

To demonstrate that the results are not a product of the small sample, I also employ a difference-in-differences design. In this research design, I compare individual legislators’ behavior across terms when they won primary elections versus terms when they lost. The advantage of this design is its ability to utilize the full sample, thereby representing a larger number of legislators and offering greater statistical power. As shown in the results, both designs yield similar conclusions, reinforcing the findings and providing robust evidence.

Specifically, I estimate the effect of winning primary election from the following specifi-

cation,

$$Y_{ipt} = \beta_0 + \beta_1 \text{PrimaryWin}_{it} + \delta_i + \gamma_{pt} + \varepsilon_{ipt}.$$

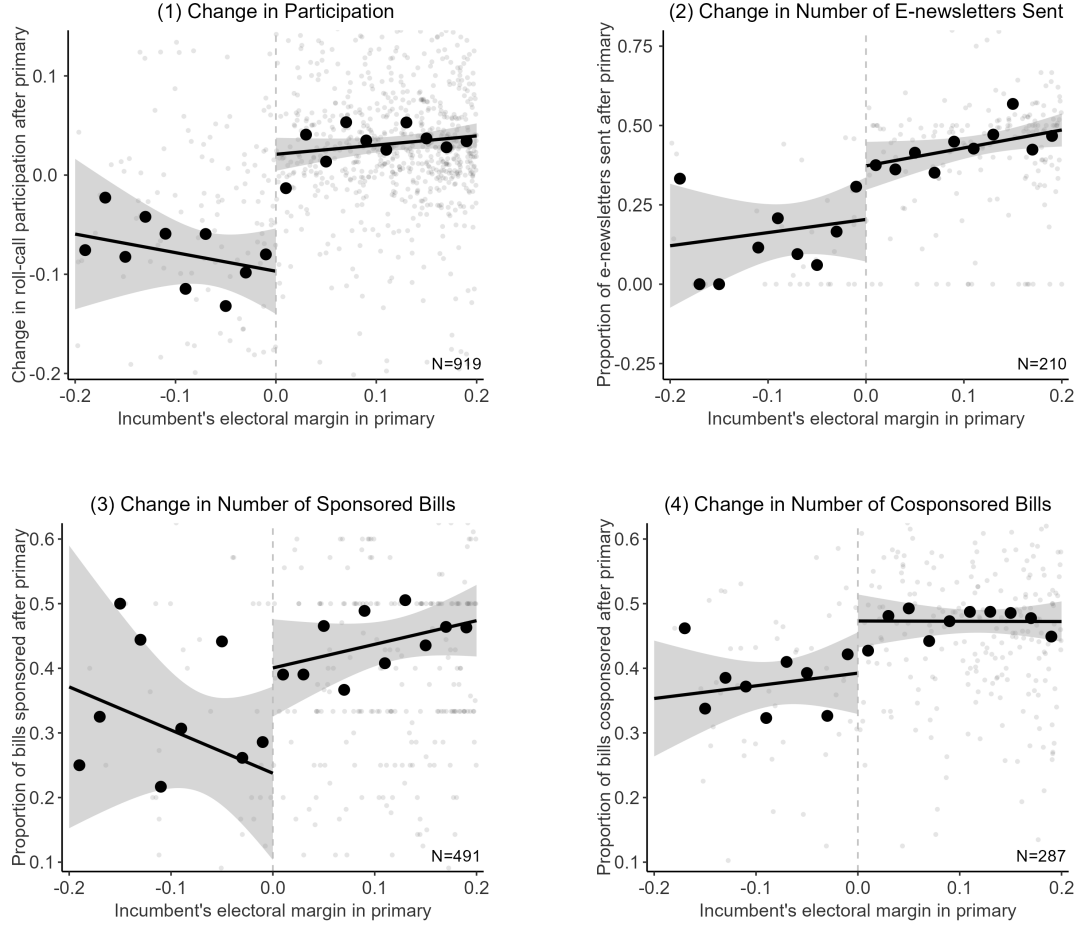
Here, PrimaryWin_{it} takes value 1 when the legislator i in Congress t won primary election and 0 otherwise. δ_i denotes legislator fixed effects, and γ_{pt} is for congress-party fixed effects. I include legislator fixed effects to account for the possibility that individual legislators may exhibit different baseline levels of behavioral change after primary elections due to personal or district-specific characteristics.³ I use congress-party fixed effects to ensure that the results are not driven by differential trends across parties and Congress. In addition to potential idiosyncrasies of each party and Congress, this fixed effect term accounts for the possibility that a legislator’s behavior may be affected by her party’s majority status. With these fixed effects terms, the coefficient of interest β_1 measures the effects of winning or losing primary elections on the legislators’ change in behavior after primary election holding fixed legislator-specific characteristics and party-congress trends.

4 Primary Losers Exhibit Decreased Effort

In this section, I show the results of both designs on legislators’ various effort-related behaviors. First, Figure 4 visualizes the discontinuous difference in the effort-related behaviors between primary election winners and losers. In each of the four graphs, the x -axis represents the running variable, or incumbents’ winning margin in primary elections, while the y -axis represents the dependent variable, or legislators’ change in behavior after the primary election. The larger black points are local means in 0.02 point bins of the x -axis, and the small gray points are raw data points. The dots to the right of the threshold, or 0, on the x -axis represent primary election winners, and those on the left represent primary election losers. The two lines are OLS fits from raw data estimated separately on each side of the threshold at 0. The shaded regions show 95% confidence intervals around the fitted lines.

³Legislator fixed effects may be unnecessary when the dependent variable is already a legislator’s within-term change in behavior. The results are not significantly different with or without legislator fixed effects.

Figure 4: Regression Discontinuity Results on Legislators' Effort-related Activities



Note: Compared to legislators who marginally lose in primary elections, those who narrowly win exhibit an increase in various effort-related activities after the primaries. In the graphs, large black points represent averages computed within 0.02-point bins of the incumbent legislators' winning margins, while small gray points show the raw data. The lines display OLS fits estimated separately on each side of the threshold, and the y-axis indicates the change in legislators' behavior before and after the primary elections.

Consider panel (1) of Figure 4, which illustrates the change in legislators' roll call participation after primary elections. The plot shows a noticeable jump at the point of discontinuity; when legislators go from barely losing to barely winning a primary election, there is a significant increase in their roll call participation rate in the post-primary periods. A similar pattern is observed in the other panels of Figure 4, showing discontinuous increases in the number of e-newsletters sent, as well as in the numbers of sponsored and cosponsored bills.

Table 3 presents numerical estimates of how winning or losing a primary election influ-

Table 3: Numerical Results on Legislators' Effort-related Activities

Regression Discontinuity				
	Change in Roll-call Participation	Change in E-Newsletters	Change in Sponsorship	Change in Cosponsorship
Primary Win	0.119*** (0.018)	0.169** (0.073)	0.163** (0.076)	0.081* (0.042)
Observations	920	210	491	287
Winner Mean	0.032	0.443	0.445	0.473
Loser Mean	-0.084	0.173	0.284	0.377
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
Difference in Differences				
	Change in Roll-call Participation	Change in E-Newsletters	Change in Sponsorship	Change in Cosponsorship
Primary Win	0.105*** (0.017)	0.201*** (0.069)	0.175** (0.073)	0.096*** (0.021)
Observations	3,696	863	2,188	1,321
Winner Mean	0.028	0.470	0.452	0.474
Loser Mean	-0.084	0.173	0.284	0.377
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The difference-in-differences results report standard errors clustered by legislator. The numbers of observations are different across columns because the data for each activity span different periods.

ences legislators' effort-related activities under both research designs. The top panel reports the coefficients from the regression discontinuity design, while the bottom panel displays those from the difference-in-differences design. Both approaches show that winning a primary election is followed by a significant increase in effort-related behavior. The alignment in the coefficient magnitudes and directions across the two designs suggests that these results are both robust and consistent.

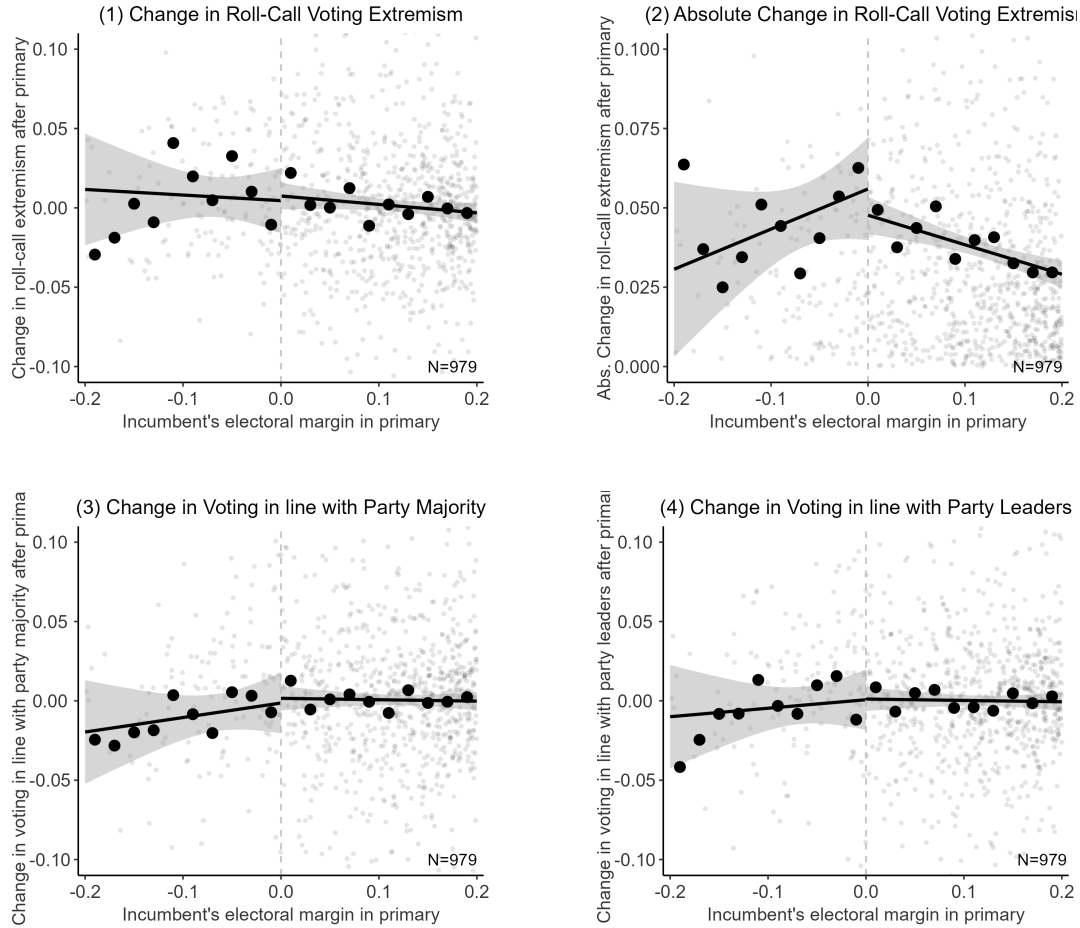
As discussed before, the change in roll call participation is measured as a regression

coefficient, representing a percentage-point change in participation rate. The results indicate that legislators who lose their primary election become approximately 10 to 11 percentage points less likely to participate in roll call votes. Changes in other effort-related behaviors are measured as post-primary proportions, where a value of 0.5 signifies no change in behavior after the primary. The table shows that losing a primary election leads to a significant decline in e-newsletter activity, bill sponsorship, and cosponsorship, as reflected in all corresponding coefficients. As shown in the winner and loser mean rows, winners' means generally fall just below 0.5 and range from 0.474 to 0.443, suggesting their behavior remains relatively stable after the primary. In contrast, primary losers' means range from 0.173 to 0.377, highlighting a substantial reduction in effort-related activities. These findings suggest that when legislators lose a primary and can no longer bid for reelection in the upcoming general election, they tend to put less effort into participating in the legislative process and communicating with their constituents.

These findings closely align with conventional literature that argues legislators' effort-related behavior is driven by their reelection motivation (Mayhew, 2004; Fenno, 1978; Cox and McCubbins, 2007). Conversely, it contrasts with predictions by scholars who contend that voters' insufficient attention and political knowledge constrain representatives' responsiveness and thus the effects of electoral incentives are limited (Achen and Bartels, 2017; Somin, 2016).

Nevertheless, a noteworthy finding is that even after losing primary elections and becoming ineligible for reelection, lame duck legislators continue to participate in most roll call votes and engage in various legislative activities, albeit at a lower frequency than their winning counterparts. After losing their primaries and until the general election, on average, these legislators still participate in 79% of roll call votes, sponsor 2.67 bills, cosponsor 49.41 bills, and send 12.33 e-newsletters. These figures indicate that electoral incentives may not be the sole driver of legislative activities. Their continued engagement may originate from ideological commitment, policy advocacy, efforts to preserve their reputation for

Figure 5: Regression Discontinuity Results on Legislators' Policy Positions



Note: Legislators who narrowly win and lose do not show a statistically significant difference in their changes in roll call voting after the primaries. In all measures, the changes for both winners and losers are statistically indistinguishable from zero, implying that neither group significantly alters their policy positions.

post-Congress careers, or a combination of these and other motivations.

5 Both Winners and Losers Show No Significant Shift in Policy Positions

This section explores whether legislators systematically alter their policy positions after losing the primary election. Similarly to the previous graphs, Figure 5 presents a visualization

of the results from the regression discontinuity design where the y -axis represents legislators' change in roll call voting patterns.

The panels (1), (3), and (4) of Figure 5 show how legislators change in roll call voting extremism, voting in line with party majority, and voting in line with party leaders, respectively. In all graphs, there appears to be a negligible shift in the trend at the threshold. This suggests that whether a legislator wins or loses close primary elections and thus runs or does not run for reelection in the upcoming election does not significantly affect their policy positions in casting roll call votes in the post-primary periods.

Table 4 presents the formal estimates, and the results from both designs consistently indicate negligible changes in policy positions. In the first, third, and fourth columns, all coefficients are at most 0.01, suggesting minimal differences in how winners and losers adjust their positions. The table also reports the mean changes in policy positions for winners and losers separately, and in every case, these mean changes are close to zero. This pattern indicates that not only do winners and losers exhibit similar changes in voting behavior following the primary elections, but either of them also barely alter their policy positions between the pre- and post-primary periods.

These results suggest the limited effects of electoral concerns on legislators' ideological positions, standing as a direct contrast with the conventional thought that position-taking is an important component of the electoral connection between legislators and their voters (Mayhew, 2004; Downs, 1965). Instead, this result stands in line with past empirical findings on the rigidity of candidate positions (Hall, 2019; Rogers, 2017; Poole and Rosenthal, 2014; Lee, Moretti and Butler, 2004) as well as formal models that assume fixed policy positions of candidates (Alesina, 1988; Osborne and Slivinski, 1996).

It is possible that changes in legislators' policy positions may not only manifest at the aggregate level. Instead, individual legislators within the same party might shift in opposing directions after primary elections, and as a result, their policy position changes may cancel each other's, thereby producing null results as a whole. To examine this possibility, the graph

Table 4: Numerical Results on Change in Legislators' Policy Positions

Regression Discontinuity				
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Win	0.003 (0.009)	−0.008 (0.006)	0.003 (0.008)	0.000 (0.008)
Observations	918	918	918	918
Winner Mean	0.001	0.037	0.001	0.000
Loser Mean	0.007	0.048	−0.001	−0.002
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
Difference in Differences				
	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Win	0.007 (0.008)	−0.014** (0.006)	0.010 (0.008)	0.002 (0.008)
Observations	3,694	3,694	3,694	3,694
Winner Mean	−0.000	0.032	−0.000	0.001
Loser Mean	0.007	0.047	−0.007	−0.003
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The difference-in-differences results report standard errors clustered by legislator. The numbers of observations are different across columns because the data for each activity span different periods.

(2) of Figure 5 and the second columns of Table 4 examine the absolute changes in roll call extremism and report slight negative coefficients, providing weak evidence that compared to primary winners, primary losers change their voting pattern slightly more frequently after primary elections. These patterns are plausible within models of electoral accountability, where legislators freed from immediate electoral pressures may shift either more extremely or more moderately according to their individual ideological preferences (Fiorina, 1973). However, the statistical significance is not robust across the two research designs and different specifications, and such an interpretation remains rather speculative.

6 Heterogeneous Effects of Electoral Incentives

6.1 Effort Effects Are More Substantial in Districts with Competitive General Elections

If primary election winners exhibit more effort-related behavior than losers after the primaries, it likely reflects their strategic preparation for the upcoming general elections, where such efforts can enhance their vote share. However, in some districts the general election is expected to be minimally competitive, especially when a legislator or their party holds a dominant position. In these cases, legislators may be less concerned with securing reelection, and primary winners in such districts might face different incentives than those in more competitive environments (Burden, 2004). This suggests that the behavioral changes observed after a primary victory could depend on the anticipated level of competition in the forthcoming general election.

To examine this possibility, I divide the primary elections in the data into two cases, one where primary election winners are expected to win the upcoming general election with relative ease to the other where primary election winners anticipate a competitive general election. Using the fact that incumbent legislators typically obtain large vote shares in general elections in uncompetitive districts while it is not the case in competitive districts, I divide the two groups using the legislators' vote margins in the most recent general elections, one with 75% point or greater vote shares and the other with less than 75% point vote shares.

Table 5 shows the comparison of the results between the two cases. The top table shows the results from the competitive districts and the bottom table shows the results from the uncompetitive districts. While primary election winners participate in roll call voting more frequently than losers in both cases, the results suggest that the positive effects of winning primary on the effort-related behavior are more substantial and robust in competitive districts. In districts with less expected competition in the general election, many of the effort-related behavior other than roll calls participation show insignificant change after pri-

Table 5: Results on Effort-related Behavior in Competitive and Uncompetitive Districts

Competitive Districts ($V_{t-1} < 0.75$)				
	Regression Discontinuity			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.081*** (0.020)	0.219*** (0.082)	0.164* (0.087)	0.089* (0.049)
Observations	655	155	368	208
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.090*** (0.020)	0.291*** (0.063)	0.252*** (0.087)	0.122*** (0.027)
Observations	2,707	635	1,600	948
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes
Uncompetitive Districts ($V_{t-1} > 0.75$)				
	Regression Discontinuity			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.217*** (0.038)	-0.330 (0.203)	0.135 (.178)	0.045 (0.081)
Observations	265	55	123	79
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.150*** (.033)	0.160*** (.044)	-0.163 (.229)	0.081 (.054)
Observations	989	228	558	373
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The top table shows the results from competitive districts where the incumbent legislator's vote share in the previous general election was less than 75%. The bottom table shows the results from less competitive districts where the previous vote share was greater than 75%. The difference-in-differences results report standard errors clustered by legislator.

mary elections or the two research designs provide inconsistent results. These results suggest that electoral incentives, rather than non-electoral factors such as the anticipation of serving another term or future committee assignments, are likely to be the primary drivers of the results.

On the other hand, the legislators’ policy positions based on roll call voting are mostly invariant in both regression discontinuity and difference-in-differences designs. Detailed results about the two groups’ position changes are in the appendix (Table A.3). These results suggest that legislators seem to primarily employ effort-based activities like sponsoring and cosponsoring bills, and sending e-newsletters to constituents to compete in the general election, whereas adjustments of policy positions are not a major strategy for electoral competition.

Notably, roll call participation coefficients are higher in uncompetitive districts than in competitive ones. A consistent explanation is that a substantial share of legislators’ motivation in attending roll calls is non-electoral – such as personal ideological commitments. This aligns with the earlier finding of this article that legislators still take part in most roll calls after losing primaries, albeit less frequently than winners. Given such a motivation, primary winners in competitive districts may attend fewer roll calls than those in uncompetitive districts because they now need to devote more time to campaigning and visiting home districts ahead of the general elections, a concern that primary losers do not share.

6.2 Effects on Policy Positions Remain Null for Important Bills

The earlier absence of evidence for shifts in policy positions might originate from the possibility that legislators only respond to electoral incentives for a small subset of bills that are salient or important. To examine the extent to which legislators’ policy positions remain consistent regardless of lame duck status, I further assess the impact of winning or losing primaries by focusing specifically on bills deemed important.

There are multiple approaches to identifying “important” bills. As a first step in exploring whether legislators respond to pressure from party leaders, I examine a subset of bills that

were “whipped” by party leadership. Party leaders occasionally conduct whip counts – informal surveys among legislators on significant legislative agendas – which are typically reserved for high-priority bills. If electoral incentives exert influence through party pressure, its effects are likely to be more pronounced for such legislation. This analysis leverages whip count data from the Democratic Party (1955–1986) and the Republican Party (1975–1980) provided by Evans (2012).

In the second approach, I focus on bills deemed especially significant or salient by previous scholars. Drawing on the dataset of priority agenda items from Curry and Lee (2020), which links these items to specific bill numbers, I replicate the same measure of position change while restricting the analysis to these priority agenda bills. If legislators are more responsive to high-profile legislation, the effects of electoral incentives may be more pronounced within this subset than across the entire range of legislative proposals.

Lastly, I assess the effects by focusing on contested bills – those that are close to passing or failing. Such bills are especially informative because each legislator’s probability of casting a pivotal vote is higher than more lopsided bills. If electoral incentives influence voting behavior by making legislators more attentive to legislative outcomes, their effects should be more pronounced on narrowly decided bills, where each legislator’s vote is more likely to be consequential. To operationalize this, I calculate the difference between yea and nay votes for each bill and repeat the analysis for bills in the lowest 30% of that difference (fewer than 62 votes difference between yea and nay counts).⁴

Each cell in Table 6 displays estimates of the effect of winning a primary election on subsequent voting behavior for “whipped,” “significant,” and “contested” bills. Overall, across both the regression discontinuity and difference-in-differences specifications, the coefficients for changes in roll-call extremism and voting alignment – whether measured against the party majority or party leaders – are small in magnitude, with most being statistically indis-

⁴30% is arbitrary but the results do not change with more or fewer bills, including lowest 20% bills or 50%. This measure of “contestedness” is a posterior measure, or a measure of the voting outcomes realized after the effects of electoral incentives could have influenced legislators’ votes. However, it is plausible that the realized contestedness is likely to be a good approximate measure of expected contestedness of the voting.

Table 6: Effects of Winning Primary on Voting on Important Bills

	Regression Discontinuity			Difference-in-Differences		
	Whipped Bills	Significant Bills	Contested Bills	Whipped Bills	Significant Bills	Contested Bills
Change in Roll Call Extremism	−0.050 (0.065)	−0.058 (0.055)	−0.006 (0.015)	−0.088** (0.041)	−0.033 (0.046)	0.002 (0.014)
Absolute Change in Roll Call Extremism	−0.037 (0.047)	0.064 (0.044)	−0.014 (.011)	0.004 (0.033)	−0.010 (0.033)	−0.023** (0.009)
Change in Voting with Party Majority	−0.034 (0.068)	0.023 (0.054)	0.001 (0.016)	0.011 (0.055)	0.003 (0.036)	0.006 (0.013)
Change in Voting with Party Leaders	−0.021 (0.065)	0.008 (0.047)	−0.025 (0.018)	0.009 (0.054)	0.027 (0.033)	−0.012 (0.013)
Change in Roll Call Participation	0.201*** (0.042)	0.186*** (.054)	0.091*** (0.017)	0.177*** (0.053)	0.192*** (0.069)	0.084*** (0.015)
Observation	323	324	843	1,250	1,363	3,483
Specification	Bandwidth = 0.2 Specification = Linear			Legislator Fixed Effects Party-Congress Fixed Effects		

Note: The number in each cell represents the coefficient of winning primary for the change in respective voting behavior, where standard errors are in the parentheses. Standard errors are clustered at the legislator level in difference-in-differences design.

tinguishable from zero. These null findings consistently indicate that primary winners and losers do not differ in how they adjust their policy positions on major legislative votes following primary elections, thereby reinforcing the previous evidence on the rigidity of legislators' policy positions.

A notable finding is that, while the effects on policy positions appear mostly null, there is a clear contrast in the stronger effects on roll call participation. For whipped and significant bills, the coefficients for roll call participation range from 0.201 to 0.177 – substantially higher than the original estimates of 0.118 under RDD and 0.105 in the difference-in-differences approach. These patterns suggest that legislators' incentives to engage in legislative activities increase markedly for high-profile legislative votes, even though this uptick in participation does not necessarily lead to meaningful shifts in policy positions.

7 Conclusion

Electoral accountability and incentives lie at the heart of democratic governance, as they compel representatives to remain responsive to their constituents. Alexander Hamilton underscored the importance of electoral incentives during the debates over term limits, arguing that removing the prospect of reelection would diminish legislators’ motivation to serve conscientiously: “One ill effect of the exclusion [of reelection] would be a diminution of the inducements to good behavior” (Hamilton, 2003).

Despite the recognized importance of electoral incentives in democratic politics, identifying precisely which “good behavior” they induce remains empirically challenging. Representatives strategically choose whether to seek reelection or retire, making it difficult to isolate the influence of electoral incentives from selection bias. This study addresses this challenge by leveraging a credible research design focused on the post-primary election period, during which incumbents face a critical juncture – continuing their reelection campaigns or exiting the race after a primary defeat. The evidence of this article shows that those still in electoral contention – primary winners – demonstrate higher levels of legislative effort compared to primary losers, including more active roll call participation, bill sponsorship, and constituent communication. On the other hand, regardless of whether they won primary election and thus anticipate to proceed to the general election, the legislators did not show significant change in their policy positions in roll call voting.

Taken together, these findings carry a range of broader implications for how scholars and practitioners understand representative behavior and related issues. First, they delineate the scope and magnitude of electoral incentives’ influence and thereby evaluate the foundational assumptions of major theoretical accounts on representative behavior, contributing to the extensive literature on electoral accountability in democratic politics. The finding that legislators exert greater effort when planning to compete in the general election reinforces classic theories of electoral accountability, indicating that the looming pressure of a general election spurs legislators to invest time and energy into their work (Barro, 1973; Ferejohn,

1986; Mayhew, 2004). Contrary to claims that voters' inattention significantly limits accountability, the results suggest that representatives do, in fact, respond to the expectation of competition by stepping up their visibility and output when reelection is on the horizon (Achen and Bartels, 2017). The observed patterns further support the claim by showing that this heightened engagement is most pronounced in districts with highly competitive general election races.

Another important takeaway is that, while the level of legislative effort is sensitive to reelection incentives, incumbents' policy positions – such as how they vote on the ideological spectrum or how often they align with their parties – remain largely stable. Whether they still face an election or not, legislators do not systematically drift toward the center or become more extreme after the primary, and the pattern remains consistent even for bills that are considered more salient and important. Contradicting with the traditional Downsian logic of electoral competition (Downs, 1965), these suggest that factors other than electoral pressure, such as personal ideological preferences or party alignment, are more likely to be decisive in shaping a legislator's roll call stance, and therefore support the view that representatives' ideological positions are mostly inflexible (Poole and Rosenthal, 2014; Hall, 2019; Lee et al., 2004; Alesina, 1988; Osborne and Slivinski, 1996).

These results extend to other relevant debates about elections and representation. In particular, the findings inform discussions of party influence (Cox and McCubbins, 2007; Krehbiel, 1993) by examining whether legislators align their roll call voting with the party line more often when they are up for reelection – a plausible scenario given that parties control key resources, such as campaign funds, which are vital for individual legislators' electoral success. Yet, by demonstrating that legislators' party-line stance remains largely consistent, the results suggest that electoral incentives are not likely to act as a conduit for party influence over individual legislators.

The stability of legislators' policy positions also speaks to the reliability of roll call-based measures of policy preferences, such as DW-NOMINATE scores (Poole and Rosenthal, 1985;

McCarty, 2016). Although legislators' voting decisions could theoretically be shaped by electoral considerations – meaning that roll call-based measures might capture not just intrinsic ideological leanings but also external political pressures – our findings indicate otherwise. Specifically, legislators' roll call patterns remain largely stable whether or not they face reelection, suggesting that these measures predominantly reflect their core, enduring ideological preferences.

These findings also contribute to the ongoing and expanding literature on political polarization, providing indirect evidence about the roots of partisan and ideological divides. Given that incumbents, whether successful in primary or not, do not significantly shift their ideological stances in the closing months of their terms, the results imply that electoral turnover – who exit or enter office – is more likely to drive congressional polarization than any repositioning by sitting legislators themselves.

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A Appendix

A.1 Post-congress Career Trajectory of Primary Losers

Table 1 in the main article presents the post-Congress career trajectories of primary election losers. It is notable that most legislators do not pursue elected office again after losing a primary election although these are ex-post realized decisions and do not directly reveal candidates’ true incentives during their lame-duck periods. To discount the potential influence of future electoral ambitions, I conduct a separate analysis excluding all primary losers who later sought an elected office. The results remain consistent, as shown in Table A.1.

A.2 Manipulation Test

The McCrary density test is used in regression discontinuity designs to check for potential manipulation of the running variable around the cutoff point (McCrary, 2008). The idea is that if individuals can manipulate the running variable, there might be a discontinuity in its density function at the cutoff, indicating possible sorting or manipulation. The test produces Figure A.1 and an estimated value of 0.179, indicating that the density of the running variable on either side of the cutoff is not significantly different. This suggests that manipulation of the running variable is unlikely.

A.3 Excluded Legislators

In the original data, many legislators either ended casting roll-call votes before primary election dates (73) or started casting roll-call votes after primary election dates (258). Because this article’s research designs require observations of House members’ behavior before and after primary elections, these legislators are not applicable and thus dropped from the analysis. To make the estimation more reliable, I excluded Congress-legislators who cast less than ten roll-call votes either before or after the primary election. To ensure that I am excluding any exceptional cases, I also excluded any legislators whose share of roll-call votes is less than

Table A.1: Results without Re-running Primary Losers

Effort Behavior				
	Regression Discontinuity			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.120*** (0.019)	0.169** (0.082)	0.162* (0.089)	0.050* (0.091)
Observations	891	203	473	275
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Participation	E-Newsletters	Sponsored Bills	Cosponsored Bills
Primary Win	0.108*** (0.019)	0.212** (0.082)	0.190** (0.088)	0.089*** (0.026)
Observations	3,667	856	2,170	1,309
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes
Policy Position				
	Regression Discontinuity			
	Extremism	Abs. Extremism	With Party	With Leader
Primary Win	0.003 (0.010)	-0.010 (0.007)	0.004 (0.009)	0.002 (0.009)
Observations	889	889	889	889
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Extremism	Abs. Extremism	With Party	With Leader
Primary Win	0.010 (.009)	-0.014 (.007)	0.009 (.009)	0.003 (.008)
Observations	3,665	3,665	3,665	3,665
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: These results are based on the sample without any House representatives who lost in primary elections but run for an election later. The results are mostly the same with the main findings.

50% out of the entire vote share cast by representatives from that district in that Congress, eliminating legislators who occupied the seat only a small fraction of the term.

A.4 Regression Discontinuity Results with Different Specifications

The choice of bandwidth in a regression discontinuity design can be arbitrary, so it is important to demonstrate that conclusions are not driven by a particular choice of the spec-

ification. Figure A.2 shows the RDD estimate over different sizes of the bandwidths. The results are rather unstable for very narrow bandwidths due to the small sample sizes.

Due to the shorter time spans of the data for e-newsletters, sponsorship, and cosponsorship, the sample sizes are limited when using narrow bandwidths. As a result, many coefficients are statistically insignificant and fluctuate with small adjustments in bandwidth sizes within these narrow ranges. Nonetheless, the signs of the coefficients generally remain largely stable and align with the main conclusion of this article. Where the sample sizes are sufficiently large, the coefficients are largely consistent with the results from the difference-in-differences design.

A.5 Regression Discontinuity Results from Redistricting Years

The regression discontinuity design only derives local average treatment effect and is not externally valid. The specific concern is that those legislators who are subject to close primary elections might be systematically and significantly different from the other legislators who face non-close primaries or no primary challenges at all. To address this concern indirectly, I implement the same research design focusing on only redistricting years, assuming that in

Figure A.1: Plot from McCrary Test

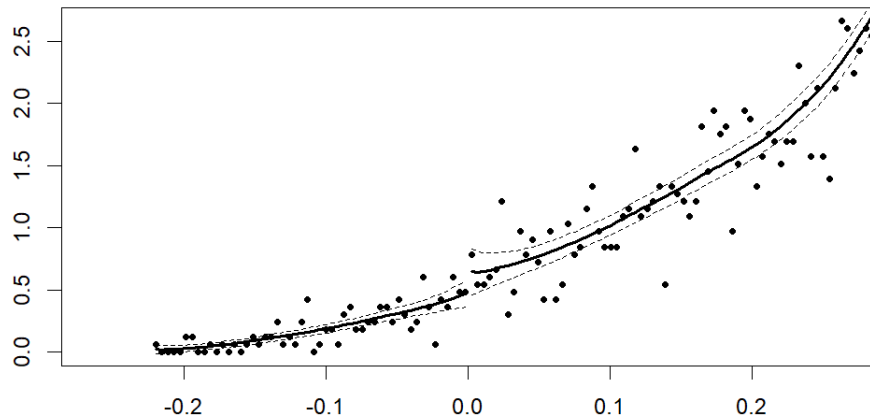
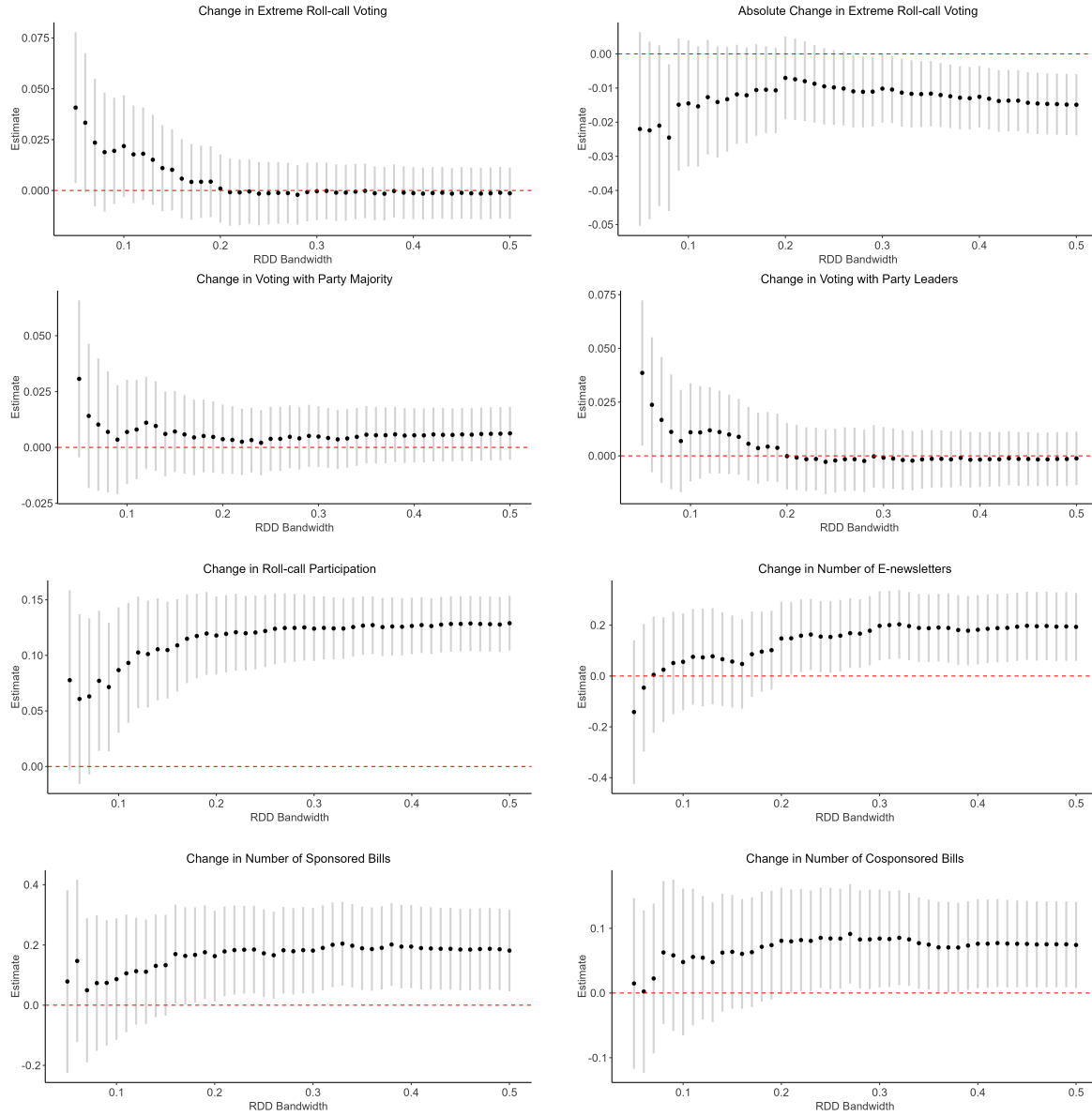


Figure A.2: RDD Estimates from Different Bandwidths



Note: Each graph visualizes the RDD estimates of changes in legislator behavior over different bandwidth. The x -axis in each graph is for the bandwidth, and the y -axis for the estimate size. The vertical gray lines for each bandwidth represents 95% confidence interval.

redistricting years, legislators are more likely to face substantial primary competition in an as-if random fashion. It suffers from fewer sample and weaker statistical power as it includes fewer election years, but the results in Table A.2 show that the sizes and direction of the coefficients remain mostly consistent with the original findings.

Table A.2: Regression Discontinuity Results from Redistricting Years

	Change in Roll-call Participation	Change in E-Newsletter	Change in Sponsorship	Change in with Cosponsorship
Primary Win	0.110*** (0.028)	0.136 (0.114)	0.052 (0.146)	0.074 (0.056)
Observations	279	73	124	111
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear

	Change in Roll-call Extremism	Absolute Change in Roll-call Extremism	Change in Voting with Party Majority	Change in Voting with Party Leaders
Primary Win	0.024 (0.015)	-0.008 (0.011)	0.011 (0.012)	0.003 (0.013)
Observations	279	279	279	279
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Standard errors are in the parentheses and are clustered by legislator.

A.6 Policy Position Change in Competitive and Uncompetitive Districts

In the main article, I assess whether the effects of effort-related behavior vary with the expected competitiveness of the general election, finding that these effects are more pronounced in districts where primary winners anticipate competitive general elections. To examine whether similar heterogeneous effects exist for legislators' policy positions, I compare changes in policy positions after the primary between competitive and uncompetitive districts.

Table A.3 shows estimates of legislators' changes in policy positions in both competitive and uncompetitive districts following the primary elections. According to the coefficients presented, neither research design detects significant differences in policy position changes between these two groups. This indicates that changes in policy positions remain unaffected

by the expected competitiveness of the upcoming general election, reinforcing the argument that adjusting policy stances is unlikely to be a predominant strategy for legislators seeking reelection.

A.7 Results Over Different Periods

An additional analysis was conducted to examine whether the effects vary across different time periods. Given the limited sample size, it was not feasible to segment the data into very fine time intervals, and I divided the sample into two periods: before and after 1994, which marks the beginning of a significant rise in ideological polarization and stricter party discipline.

The results in Table A.4 show that in both period, the main conclusions are not much different: primary losers show decline in effort-related behavior while both winners and losers show negligible changes in policy positions.

A.8 Legislators' Anticipation of Presidential Appointment

One potential consideration for primary losers is the prospect of being appointed to federal government positions or local offices, such as positions on regional committees, by the president (Jenkins and Nokken, 2008*a*). In fact, historical records show that presidents have appointed many primary election losers to these positions. Anticipating these opportunities, primary losers might adjust their behavior – for example, by closely adhering to the party line – to improve their chances of securing a desirable public appointment, or in exchange for one.

To investigate this possibility, I analyze whether electoral incentives influence policy positions differently for legislators of the president's party during midterm elections. Because the president will remain in office for at least two more years and rarely appoints legislators from the opposing party, these lawmakers are more likely than others to anticipate a post-Congress appointment. If such anticipation is a strong motivator, primary losers would

have an incentive to maintain positions aligned with the president. If the president and the party share aligned policy preferences, these losing legislators are expected to exhibit smaller changes in their policy positions compared to primary winners.

As shown in Table A.5, the results are mixed and do not clearly support the idea that these legislators will show smaller changes in policy positions after primary elections. The regression discontinuity approach reveals significant effects on legislators' policy positions: compared to primary losers, primary winners are 3.4 percentage points more likely to cast an extreme roll-call vote and 3.5 percentage points more likely to vote in line with party leaders. In contrast, the difference-in-differences design yields minimal and statistically insignificant differences between winners and losers. Overall, the mixed findings do not provide strong evidence that the anticipation of a presidential appointment plays a crucial role in shaping primary losers' post-primary behavior.

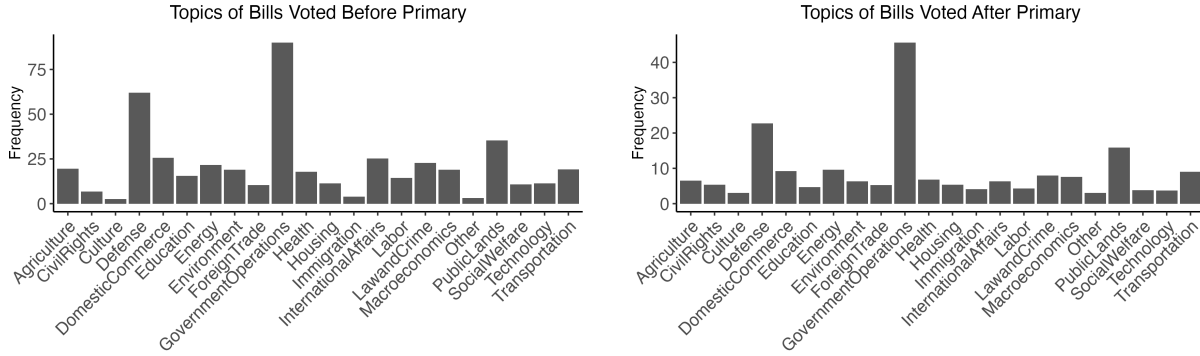
A.9 Descriptive Bill Differences Before and After Primary Elections

In the main article, I account for potential differences in the types of bills voted before and after primary elections by using regression models with bill fixed effects, thereby isolating changes in legislators' voting behavior from potential variations in bill content.

Nevertheless, to ensure that the composition of bills is not dramatically different before and after primary elections, I conduct several descriptive comparisons, examining the distribution of bill topics and legislators' general voting patterns.

Figure A.3 presents the average number of topics of bills voted on per congressional term before (left) and after (right) primary elections. The topics are classified according to the Congressional Bill Project (Adler and Wilkerson, 2020). The distributions appear largely consistent across both periods, with government operations being the most frequently addressed topic, followed by defense, public lands, and other policy areas in a similar order. These patterns suggest that the legislative agenda remains relatively stable before and after

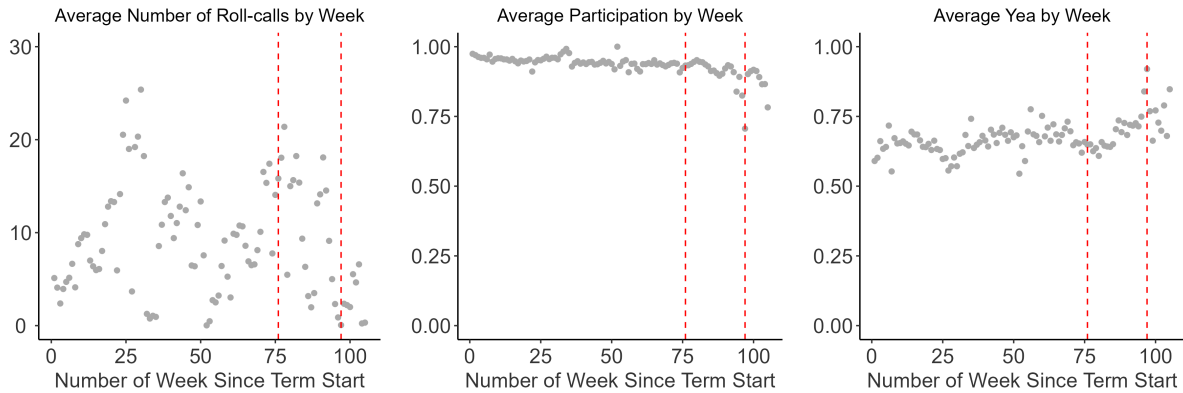
Figure A.3: Distribution of Topics of Bills Before (left) and After Primary (right)



primary elections.

Figure A.4 illustrates three aggregate characteristics of bills that reach a roll-call vote across a congressional term: the number of bills voted on per week (left), legislators' participation rates in those votes (center), and the frequency with which bills pass (right). The two red vertical lines indicate the average dates of the primary election and the general election, serving as reference points to assess whether the nature of bills considered changes before and after primary elections. While the plots reveal natural week-to-week fluctuations and a sharp shift near the general election, there is no clear evidence of a systematic or substantial

Figure A.4: Average Weekly Roll-call Voting Characteristics



Note: The x-axis of each figure represents the number of weeks since the start of a Congress. The first figure plots the average number of roll-call votes per week, the second displays the average participation rate, and the third shows the average yea vote rate. The two red vertical dotted lines indicate the time window between the average primary election date and the general election.

change in bill characteristics around the primary election period.

A.10 Policy Position Results by Mean Differences Between Pre- and Post-Primary Periods

In the main article, I assess legislators' changes in roll-call voting using a regression approach, estimating them through OLS coefficients. As a robustness check, I additionally measure changes in legislators' voting behavior using a simpler method – calculating the mean difference in their voting patterns between the pre- and post-primary periods. This alternative approach ensures that the results are not driven by the specific estimation method used in the main analysis.

Specifically, for each legislator i in t , I measure a legislator's change in voting pattern after primary election by the difference of average voting patterns in the pre- and post-primary periods. For example, if legislator i in t , participated 90% and 70% of roll-calls in the pre-primary period and post-primary period, respectively, then the change is 20% point decrease and I denote it as $Y_{it} = -0.2$.

In the second analysis, to ensure that the time windows are directly comparable, I restrict the sample to roll-calls occurring within the same specific time window in each year. For instance, if a primary election is held on August 1 and the general election on November 1, I compare the participation rate from August 1 to November 1 in the election year with the rate during that same period in the preceding year. As shown in Table A.6, the results remain largely consistent in both specifications.

Table A.3: Changes in Policy Positions in Competitive and Uncompetitive Districts

Competitive Districts ($V_{t-1} < 0.75$, $N = 2,706$)				
	Regression Discontinuity			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.005 (0.010)	-0.000 (0.007)	0.006 (0.009)	0.000 (0.009)
Observations	653	653	653	653
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.004 (0.009)	0.000 (0.006)	0.011 (0.007)	-0.001 (0.007)
Observations	2,705	2,705	2,705	2,705
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes
Uncompetitive Districts ($V_{t-1} > 0.75$, $N = 989$)				
	Regression Discontinuity			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.000 (0.019)	-0.038*** (0.014)	-0.001 (0.017)	0.008 (0.017)
Observations	265	265	265	265
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.016 (0.017)	-0.028* (0.014)	-0.009 (0.023)	-0.000 (0.017)
Observations	989	989	989	989
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The top table shows the results from competitive districts where the incumbent legislator's previous vote share was less than 75%. The bottom table shows the results from less competitive districts where the previous vote share was greater than 75%.

Table A.4: Policy Position Results in the Periods Before and After 1994

After 1994 ($N = 1,566$)				
	Regression Discontinuity			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	-0.009 (0.008)	-0.000 (0.006)	0.000 (0.009)	0.001 (0.008)
Observations	654	654	654	654
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	-0.004 (0.007)	-0.006* (0.003)	0.014** (0.006)	0.009 (0.006)
Observations	1,566	1,566	1,566	1,566
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes
Before 1994 ($N = 2,128$)				
	Regression Discontinuity			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.010 (0.013)	-0.013 (0.009)	0.005 (0.012)	0.001 (0.012)
Observations	558	558	558	558
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.020 (0.017)	-0.027** (0.014)	-0.005 (0.021)	-0.000 (0.016)
Observations	2,128	2,128	2,128	2,128
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: Comparison on the changes in policy position in the periods before and after 1994. The top table shows the results from years from 1994. The bottom table shows the results from years prior to 1994.

Table A.5: Policy Positions Among Legislators in the President's Party in Midterm Years

	Regression Discontinuity			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.034* (0.019)	-0.037*** (0.014)	0.031 (0.019)	0.035** (0.018)
Observations	226	226	226	226
Bandwidth	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear
	Difference in Differences			
	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	-0.015 (0.043)	-0.017 (0.029)	-0.007 (0.037)	0.009 (0.033)
Observations	889	889	889	889
Legislator FE	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: These results are from legislators who face primary elections during midterm election years and belong to the president's party. These legislators are more likely than others to anticipate a post-Congress appointment by the president, as he will remain in office for at least two more years and rarely appoints legislators from the opposing party.

Table A.6: Result on Policy Position by Mean Difference Before and After Primary Elections

Difference in Mean Before and After Primary					
	Regression Discontinuity				
	Change in Roll-Call Participation	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.106*** (0.018)	0.015 (0.012)	-0.021** (0.008)	0.008 (0.009)	0.012 (0.009)
Observations	918	918	918	918	918
Bandwidth	0.2	0.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear	Linear
	Difference in Differences				
	Change in Roll-Call Participation	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.090*** (0.016)	0.010 (0.009)	-0.021*** (0.007)	0.012 (0.009)	0.007 (0.008)
Observations	3,694	3,694	3,694	3,694	3,694
Legislator FE	Yes	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes	Yes
Difference in Mean Before and After Primary in Same Time Windows					
	Regression Discontinuity				
	Change in Roll-Call Participation	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.102*** (0.020)	0.008 (0.014)	-0.015 (0.010)	-0.005 (0.011)	-0.009 (0.011)
Observations	913	913	913	913	913
Bandwidth	0.2	.2	0.2	0.2	0.2
Specification	Linear	Linear	Linear	Linear	Linear
	Difference in Differences				
	Change in Roll-Call Participation	Change in Roll-Call Extremism	Absolute Change in Roll-Call Extremism	Change in Voting With Party Majority	Change in Voting With Party Leaders
Primary Win	0.098*** (0.017)	0.009 (0.010)	-0.021*** (0.007)	0.005 (0.011)	-0.003 (0.011)
Observations	3,679	3,679	3,679	3,679	3,679
Legislator FE	Yes	Yes	Yes	Yes	Yes
Party-Congress FE	Yes	Yes	Yes	Yes	Yes

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Note: The table presents estimates of the difference in legislators' roll-call voting behavior before and after primary elections. The top panel compares all votes cast after the primary to those cast before. The bottom panel refines the comparison by examining votes cast post-primary against those from the same time window in the preceding year.

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